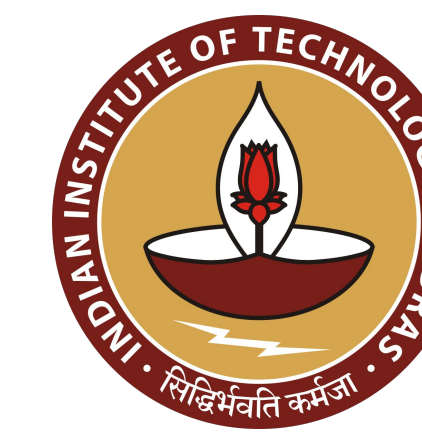


MedAIScout: Automated Retrieval of Known AI/ML Vulnerabilities in Medical Applications



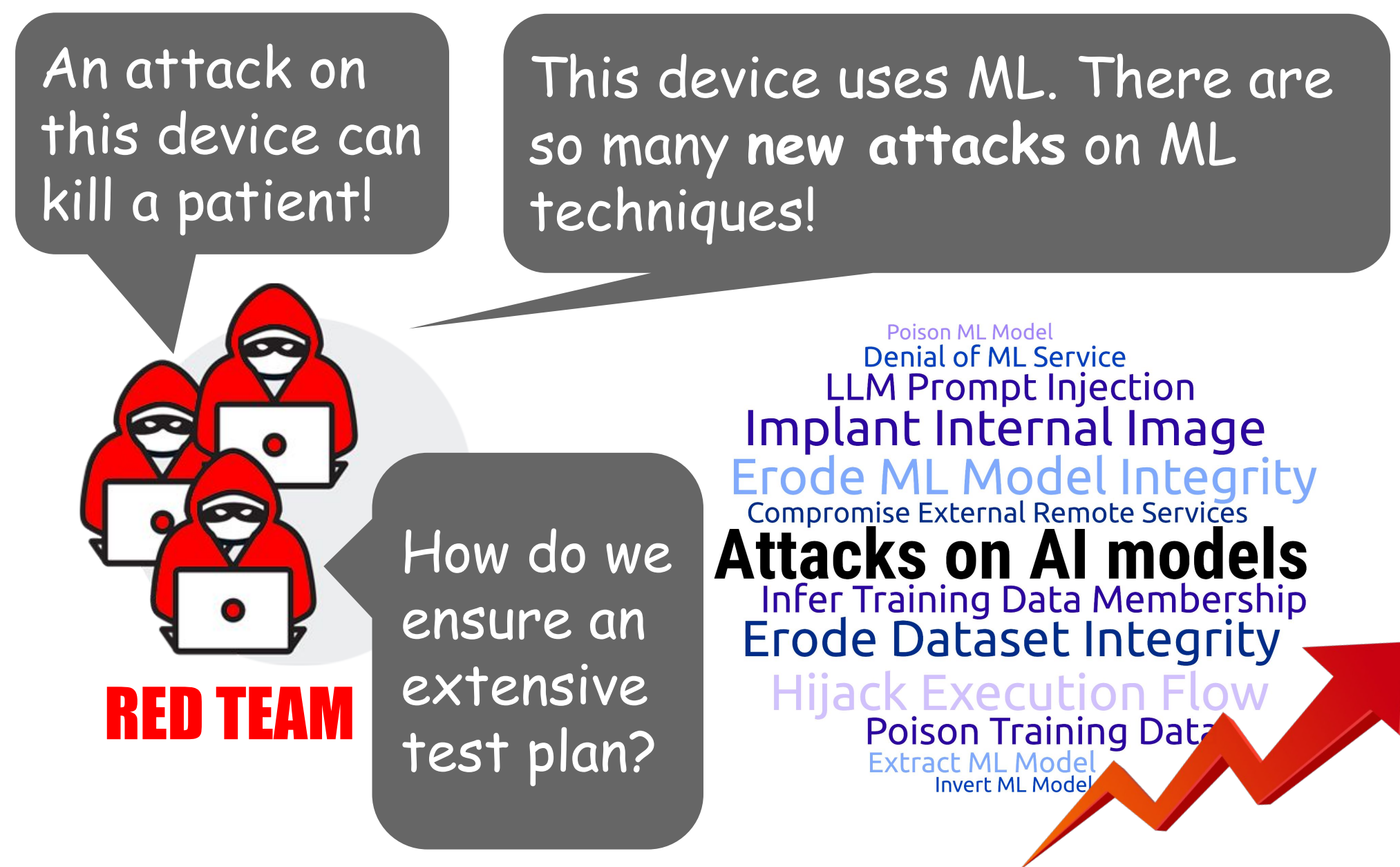
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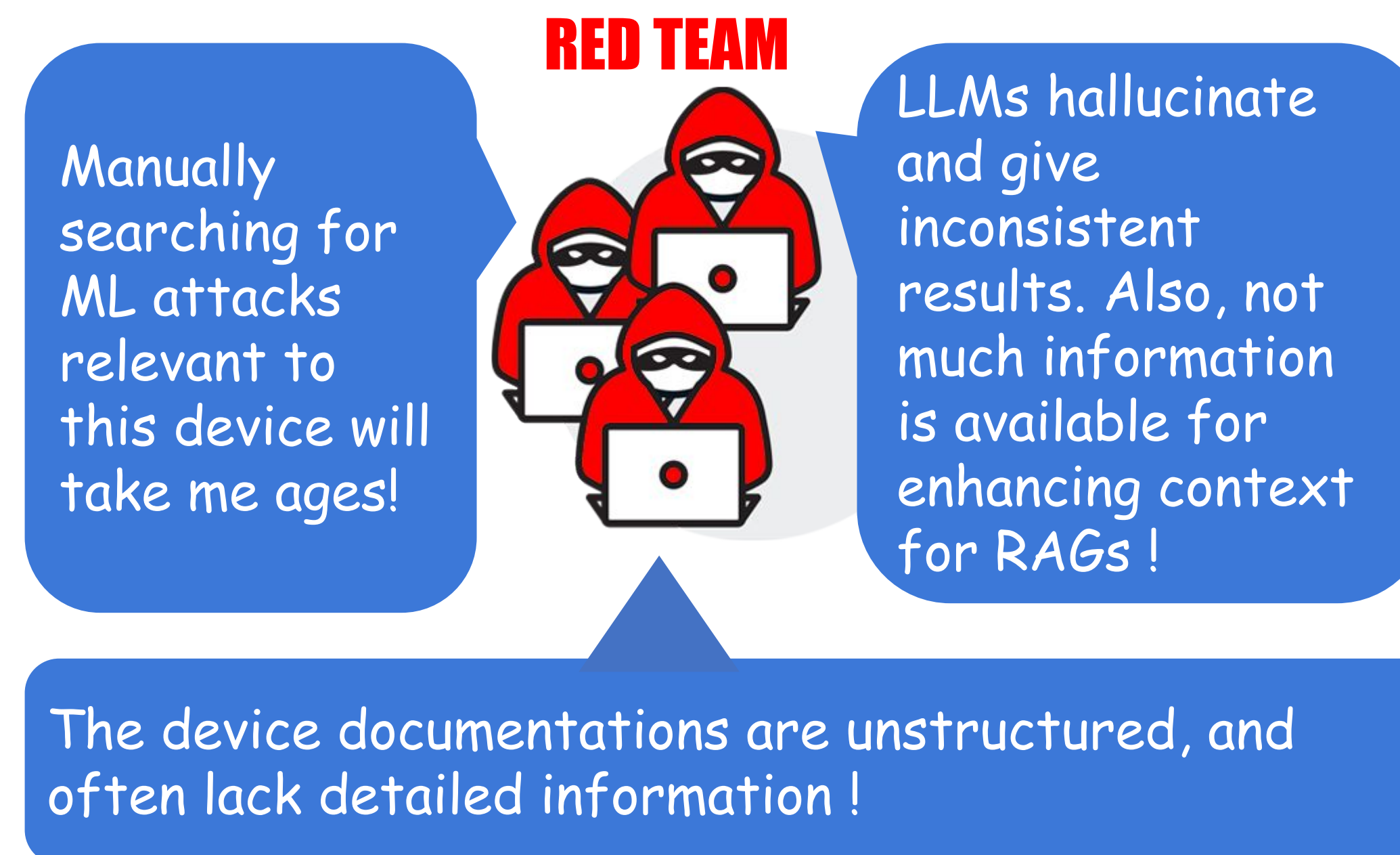
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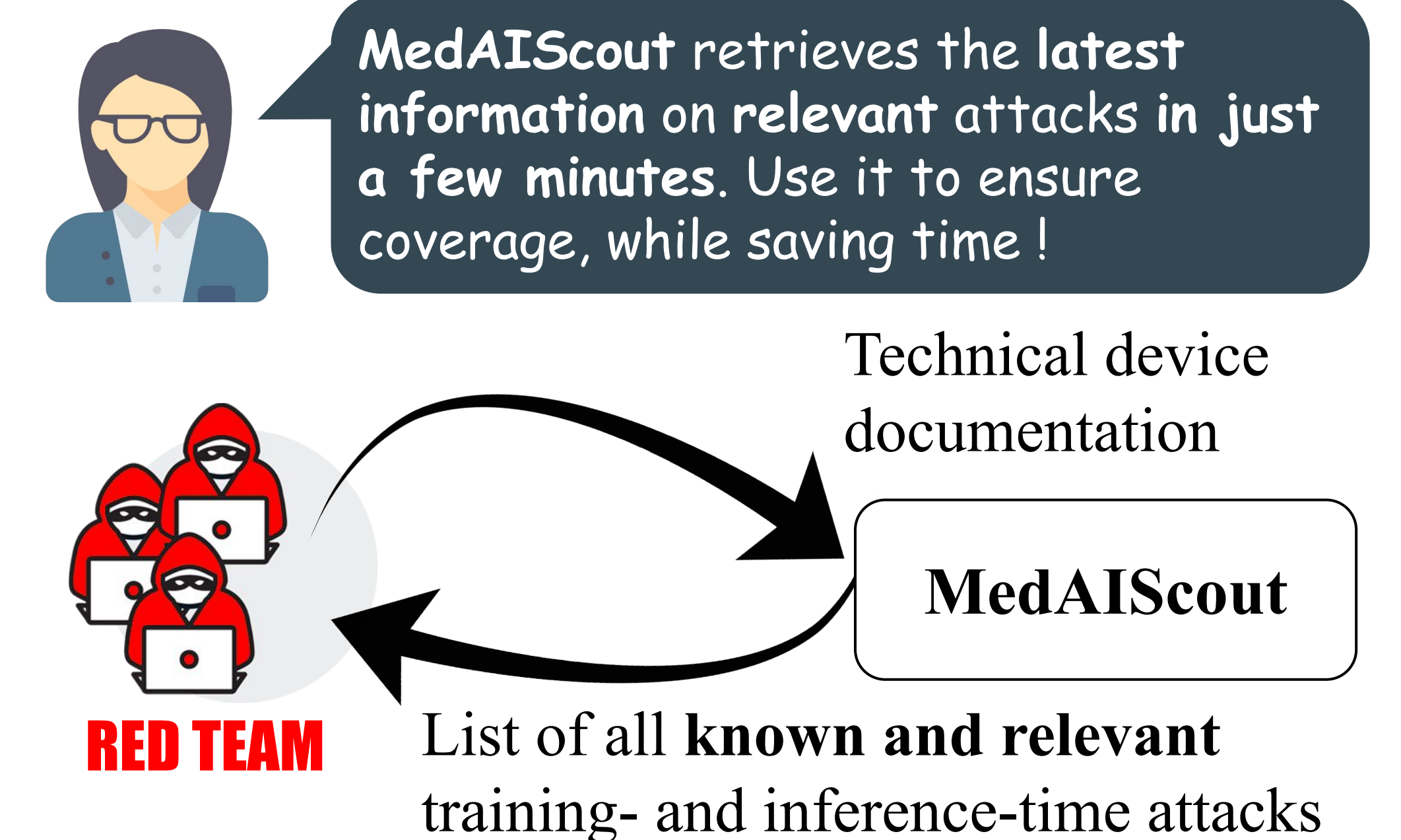
Medical AI Security: A Red Team's Race Against the Clock



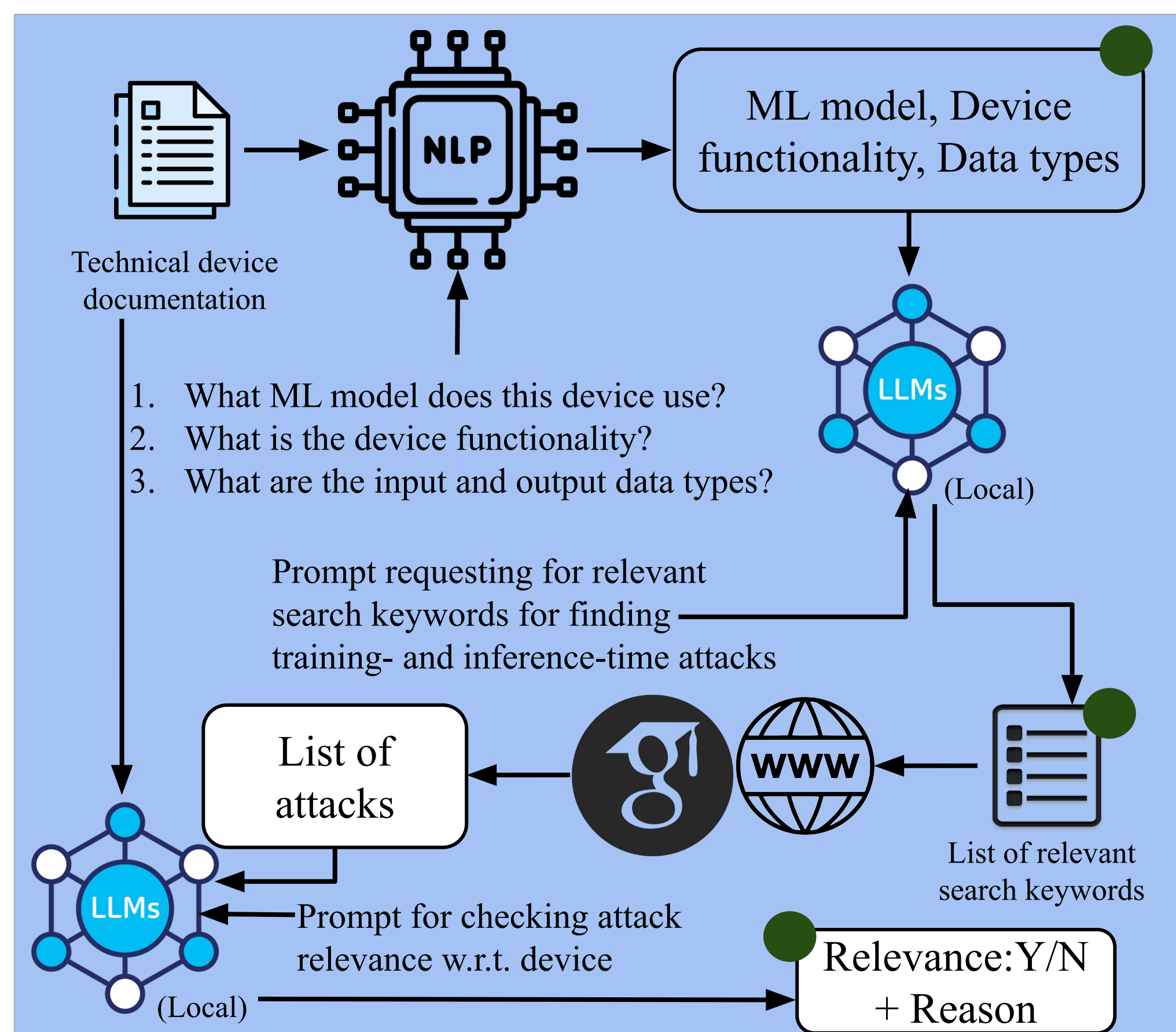
LLM and RAG: Band-Aids, Not Breakthroughs



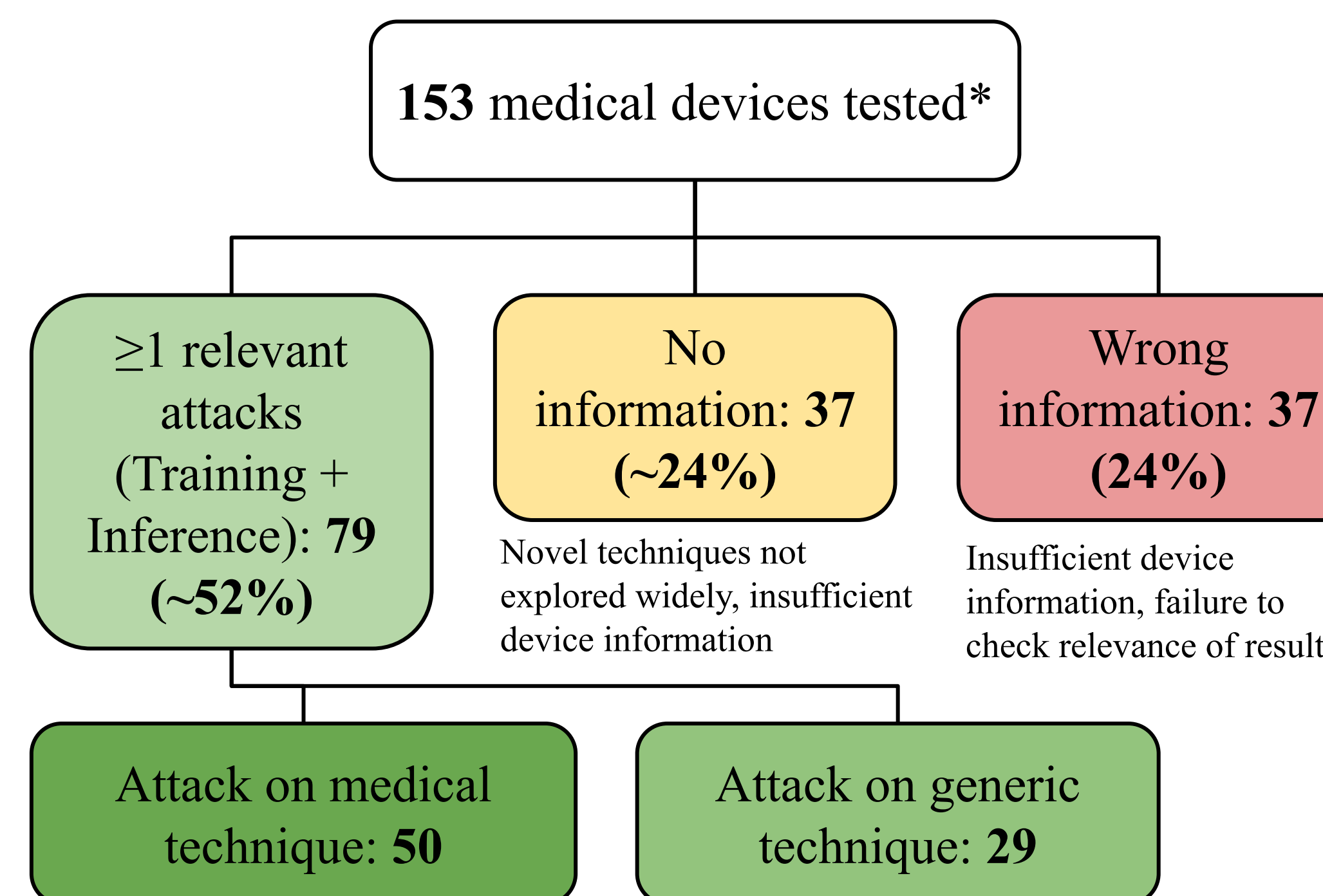
MedAIScout: Uncovering the Latest ML Vulnerability Insights



MedAIScout: What's Inside



Preliminary Experimental Results



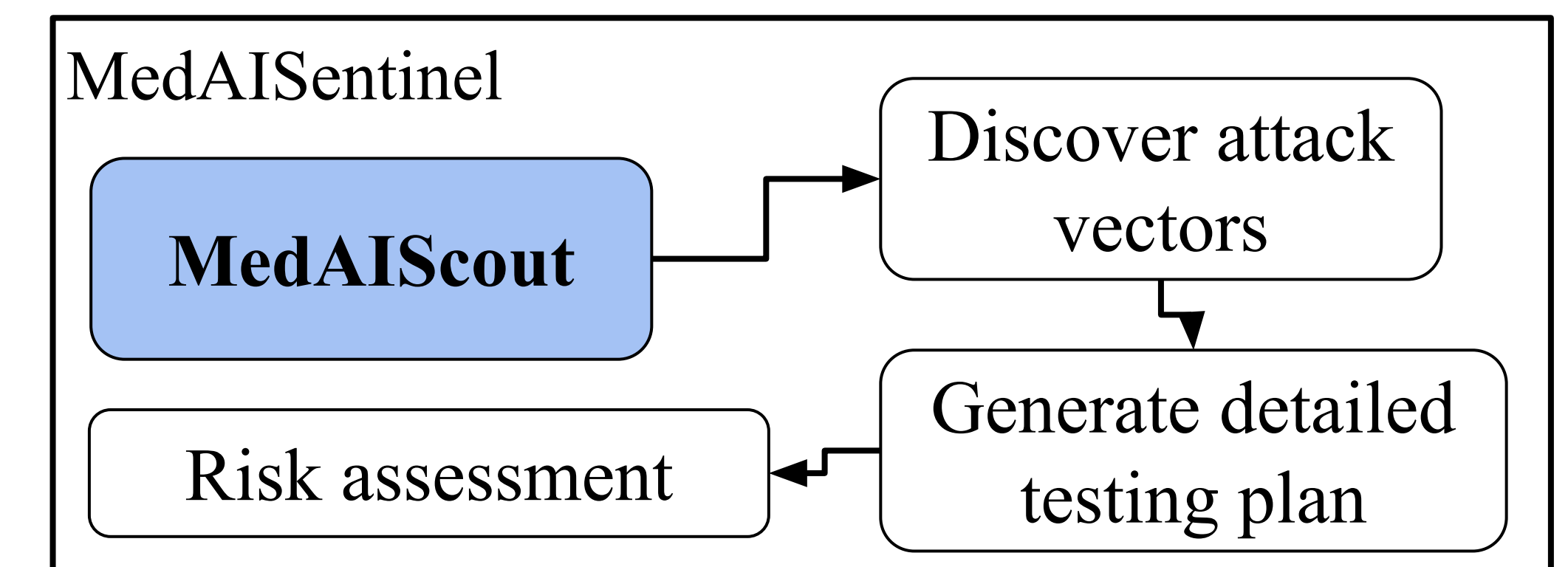
Widely explored technique, MedAIScout performs well even when device documentations lacks some technical details.

Refer to the paper for 5 detailed case studies.

* Devices were selected randomly from list of FDA-cleared devices
Documentation: Device summaries submitted to FDA + Medical Futurist
Attacks: Peer-reviewed articles from Google Scholar

The Way Ahead

1. Eliminating the need for manual validation
2. Incorporate diverse sources of information - device description + attack information
3. Strengthen relevance checking technique



References

1. U.S. Food and Drug Administration. Artificial Intelligence and Machine Learning (AI/ML)-Enabled Medical Devices. <https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-ai-ml-enabled-medical-devices>.

